Taylor and Maclaurin Series

Section 1: Common Maclaurin Series ($e^x, \sin x, \cos x$)

1 What is the general term in the Maclaurin series for $\sin x$?

a. $\frac{(-1)^n x^{2n+1}}{(2n+1)!}$

b. $\frac{(-1)^n x^{2n}}{(2n)!}$

c. $\frac{x^n}{n!}$

d. $\frac{x^{2n+1}}{(2n+1)!}$

2 What is the coefficient of x^4 in the Maclaurin series for e^{-2x}?

a. $\frac{2}{3}$

b. $\frac{16}{24} = \frac{2}{3}$

c. $-\frac{2}{3}$

d. $\frac{16}{4}$

3 Which series equals $\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n}}{(2n)!}$?

a. $\cos(\pi) = -1$

b. $\sin(\pi) = 0$

c. e^{π}

d. 1

Section 2: Constructing Taylor Polynomials from Derivatives

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4 Which formula gives the n-th degree Taylor polynomial $T_n(x)$ centered at $x = c$?

a. $\sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k$

b. $\sum_{k=0}^{n} f^{(k)}(c)(x-c)^k$

c. $\sum_{k=0}^{n} \frac{f(c)}{k!}(x-c)^k$

d. $\frac{f^{(n)}(c)}{n!}(x-c)^n$

5 Find the 2nd degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at $c = 4$.

a. $2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$

b. $2 + \frac{1}{4}(x-4) + \frac{1}{32}(x-4)^2$

c. $2 + \frac{1}{2}(x-4) - \frac{1}{8}(x-4)^2$

d. $2 + \frac{1}{4}(x-4)$

6 If a function f has $f(2) = 5, f'(2) = -3, f''(2) = 8$, what is $T_2(2.1)$?

a. 4.74

b. 4.70

c. 4.82

d. 5.04

Section 3: Manipulating Known Taylor Series

7 What is the first non-zero term of the Maclaurin series for $f(x) = x^2\cos(3x)$?

a. x^2

b. $3x^2$

c. $-x^2$

d. $\frac{9x^4}{2}$

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8 Find the Maclaurin series for $\frac{\sin x - x}{x^3}$.

a. $-\frac{1}{6} + \frac{x^2}{120} - \dots$

b. $\frac{1}{6} - \frac{x^2}{120} + \dots$

c. $1 - \frac{x^2}{6} + \dots$

d. $-\frac{x^3}{6} + \dots$

9 Evaluate $\lim_{x \to 0} \frac{\cos x - 1 + \frac{x^2}{2}}{x^4}$.

a. $\frac{1}{24}$

b. $-\frac{1}{24}$

c. 0

d. $\frac{1}{12}$

Section 4: Lagrange Error Bound (Taylor's Remainder Theorem)

10 What is the Lagrange Error Bound formula for $|R_n(x)| = |f(x) - P_n(x)|$ centered at c?

a. $|R_n(x)| \le \frac{M}{(n+1)!}|x - c|^{n+1}$, where $M = \max|f^{(n+1)}(t)|$

b. $|R_n(x)| \le \frac{M}{n!}|x - c|^n$

c. $|R_n(x)| \le |f^{(n+1)}(c)|$

d. $|R_n(x)| \le \frac{1}{(n+1)!}$

11 If $P_3(x)$ approximates $\sin x$ at $x = 0.5$ centered at 0, what is the Lagrange Error Bound using $M = 1$?

a. $\frac{(0.5)^4}{24} = \frac{1}{384}$

b. $\frac{(0.5)^3}{6}$

c. $\frac{1}{24}$

d. $\frac{(0.5)^4}{120}$

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12 On an AP free-response question, which derivative is bounded to establish the Lagrange error bound for a 4th-degree Taylor polynomial?

a. The 5th derivative $f^{(5)}(x)$

b. The 4th derivative $f^{(4)}(x)$

c. The 1st derivative $f'(x)$

d. The 6th derivative $f^{(6)}(x)$

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